

01/04/2024

Phase-I
CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 4

Time : 3 Hrs. 20 Mins.

Answers

1. (2)	41. (1)	81. (1)	121. (2)	161. (2)
2. (3)	42. (2)	82. (3)	122. (3)	162. (3)
3. (3)	43. (3)	83. (1)	123. (4)	163. (1)
4. (2)	44. (3)	84. (1)	124. (4)	164. (2)
5. (3)	45. (4)	85. (4)	125. (3)	165. (3)
6. (1)	46. (1)	86. (2)	126. (2)	166. (3)
7. (2)	47. (4)	87. (3)	127. (1)	167. (1)
8. (1)	48. (4)	88. (1)	128. (4)	168. (2)
9. (4)	49. (4)	89. (4)	129. (1)	169. (4)
10. (3)	50. (4)	90. (2)	130. (4)	170. (2)
11. (3)	51. (3)	91. (1)	131. (2)	171. (2)
12. (3)	52. (3)	92. (3)	132. (2)	172. (1)
13. (2)	53. (4)	93. (1)	133. (4)	173. (3)
14. (2)	54. (4)	94. (3)	134. (2)	174. (3)
15. (1)	55. (3)	95. (3)	135. (3)	175. (3)
16. (3)	56. (1)	96. (2)	136. (4)	176. (1)
17. (4)	57. (4)	97. (4)	137. (3)	177. (3)
18. (1)	58. (4)	98. (1)	138. (2)	178. (4)
19. (2)	59. (3)	99. (2)	139. (3)	179. (1)
20. (1)	60. (2)	100. (1)	140. (1)	180. (3)
21. (3)	61. (4)	101. (3)	141. (1)	181. (4)
22. (2)	62. (2)	102. (3)	142. (2)	182. (1)
23. (1)	63. (4)	103. (2)	143. (4)	183. (2)
24. (2)	64. (1)	104. (3)	144. (2)	184. (4)
25. (4)	65. (4)	105. (3)	145. (4)	185. (3)
26. (2)	66. (3)	106. (2)	146. (3)	186. (2)
27. (2)	67. (3)	107. (1)	147. (1)	187. (1)
28. (1)	68. (4)	108. (3)	148. (1)	188. (3)
29. (2)	69. (3)	109. (3)	149. (1)	189. (3)
30. (3)	70. (4)	110. (1)	150. (3)	190. (1)
31. (1)	71. (4)	111. (3)	151. (1)	191. (1)
32. (4)	72. (2)	112. (3)	152. (2)	192. (4)
33. (3)	73. (1)	113. (3)	153. (4)	193. (3)
34. (2)	74. (2)	114. (3)	154. (2)	194. (2)
35. (4)	75. (4)	115. (1)	155. (3)	195. (3)
36. (2)	76. (1)	116. (2)	156. (1)	196. (4)
37. (3)	77. (3)	117. (3)	157. (4)	197. (2)
38. (3)	78. (3)	118. (1)	158. (4)	198. (3)
39. (1)	79. (4)	119. (1)	159. (3)	199. (2)
40. (4)	80. (3)	120. (2)	160. (2)	200. (2)



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Answers and Solutions

PHYSICS

SECTION - A

1. Answer (2)

Work done = Area under P-V curve

$$= \frac{1}{2} \times (7 - 3) \times (500 - 150) = 700 \text{ J}$$

2. Answer (3)

$$E = \frac{3}{2}KT \Rightarrow E \propto T$$

$$T_i = 127 + 273 = 400 \text{ K}$$

$$T_f = 3 \times 400 \text{ K} = 1200 \text{ K}$$

$$T_f = 1200 - 273 = 927^\circ\text{C}$$

3. Answer (3)

If work is done by the system its volume may increase and in isothermal process the internal energy of an ideal gas remains constant.

4. Answer (2)

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \Rightarrow v_{\text{rms}} \propto \sqrt{T}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v}{v_2} = \sqrt{\frac{100}{400}} = \frac{1}{2}$$

$$v_2 = 2v$$

5. Answer (3)

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta U = \Delta Q - \Delta W$$

$$= 1200 - 300$$

$$\Delta U = 900 \text{ W}$$

6. Answer (1)

$$f = \sqrt{2\pi d^2 n v}$$

$$f \propto n$$

$$\frac{f_1}{f_2} = \frac{4 \times 10^{20}}{16 \times 10^{20}} = \frac{1}{4}$$

$$\frac{f_1}{f_2} = 0.25$$

7. Answer (2)

$$\Delta Q = 0$$

$$\Delta U = -W$$

$$= -\left(-\frac{1}{2} \times (60 + 20) \times 10^3 \times (150) \times 10^{-6}\right)$$

$$= 40 \times 10^3 \times 150 \times 10^{-6}$$

$$\Delta U = 6 \text{ J}$$

8. Answer (1)

$$PV = nRT$$

$$\frac{P}{T} = \frac{nR}{V}$$

Slope of P-T graph is inversely proportional to V.
So $V_1 > V_2$.

9. Answer (4)

$$\text{Total energy} = \frac{1}{2}kA^2$$

$$\text{Potential energy} = \frac{1}{2}k\left(\frac{A}{3}\right)^2$$

$$= \frac{1}{2} \frac{kA^2}{9} = \frac{kA^2}{18}$$

$$\text{Kinetic energy} = \frac{1}{2}kA^2 - \frac{kA^2}{18}$$

$$= \frac{9kA^2 - kA^2}{18} = \frac{8kA^2}{18}$$

$$\frac{\text{Potential energy}}{\text{Kinetic energy}} = \frac{1}{8}$$

10. Answer (3)

Time taken to go from $x = 0$ to $\frac{A}{2}$

$$\frac{T}{12} = 3 \Rightarrow T = 36 \text{ s}$$

Time taken from $\frac{A}{2}$ to A is

$$\frac{T}{6} = \frac{36}{6} = 6 \text{ s}$$

11. Answer (3)

$$x = A \sin \omega t \quad \dots(1)$$

$$v = A \omega \cos \omega t \quad \dots(2)$$

$$a = -\omega^2 A \sin \omega t \quad \dots(3)$$

From (2) & (3)

$$\frac{v^2}{(A\omega)^2} + \frac{a^2}{(\omega^2 A)^2} = 1$$

12. Answer (3)

$$\text{Amplitude} = 8\sqrt{(1)^2 + (\sqrt{3})^2} = 16 \text{ cm}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{5\pi} = \frac{2}{5} \text{ s} = 0.4 \text{ s}$$

13. Answer (2)

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{50}{5 \times 10^{-3}}} = 100 \text{ m/s}$$

14. Answer (2)

$$\text{Velocity of wave} = \frac{\text{Coefficient of } t}{\text{Coefficient of } x}$$

$$= \left(\frac{5}{20}\right) = \frac{1}{4} = 0.25 \text{ m/s}$$

15. Answer (1)

$$\frac{v}{5.06} - \frac{v}{5.14} = \frac{60}{15}$$

$$v = \frac{60}{15} \times \frac{5.06 \times 5.14}{0.08}$$

$$v = 1300.42 \text{ m/s}$$

16. Answer (3)

$$f \propto \sqrt{T}$$

$$\frac{f_A}{f_B} = \sqrt{\frac{T_A}{T_B}}$$

$$\frac{200}{150} = \sqrt{\frac{T_A}{T_B}}$$

$$\frac{T_A}{T_B} = \frac{16}{9}$$

17. Answer (4)

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Comparing this with standard equation of SHM i.e.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$\omega^2 = 4 \Rightarrow \omega = 2$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ s}$$

18. Answer (1)

$$C = C_V + \frac{R}{1-x}$$

$$C = \frac{3}{2}R + \frac{R}{1-3} \quad [\because PV^n = \text{constant so } n = 3]$$

$$C = \frac{3}{2}R - \frac{R}{2} = R$$

19. Answer (2)

$$\text{Case 1 : } PV = nRT$$

$$\text{Case 2 : } 2P\left(\frac{V}{2}\right) = nRT_f$$

$$\Rightarrow T_f = T$$

20. Answer (1)

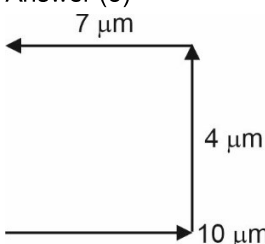
$$T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$$

$$T = 2\pi \sqrt{\frac{0.5}{\sqrt{g^2 + 3g^2}}}$$

$$= 2\pi \sqrt{\frac{0.5}{2g}}$$

$$= \frac{\pi}{\sqrt{g}} \approx 1 \text{ sec}$$

21. Answer (3)



$$\Rightarrow A = \sqrt{4^2 + (10-7)^2} = 5 \mu\text{m}$$

22. Answer (2)

In SHM, both velocity and acceleration changes continuously.

23. Answer (1)

$$f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow f^2 \propto T$$

24. Answer (2)

$$PV = \frac{m}{M}RT$$

$$P = \left(\frac{m}{V}\right) \frac{RT}{M}$$

$$\text{or } \rho = \frac{PM}{RT}$$

For a given density,

$$P_2 > P_1$$

$$\text{So } T_2 > T_1$$

25. Answer (4)

$$v = \sqrt{\frac{B}{\rho}}$$

$$v \propto \frac{1}{\sqrt{\rho}} \text{ and } v \propto \sqrt{B}$$

26. Answer (2)

$$\text{At } V = V_0, P = \frac{P_0}{2}$$

$$\therefore T_i = \frac{PV}{nR} = \frac{\left(\frac{P_0}{2}\right)(V_0)}{R} = \frac{P_0 V_0}{2R} \quad [n = 1]$$

$$\text{and at } V = 2V_0, P = \frac{4P_0}{5}$$

$$\therefore T_f = \frac{PV}{nR} = \frac{\left(\frac{4P_0}{5}\right)(2V_0)}{R} = \frac{8P_0 V_0}{5R}$$

$$\therefore \Delta T = T_f - T_i = \left[\frac{8}{5} - \frac{1}{2}\right] \frac{P_0 V_0}{R} = \frac{11P_0 V_0}{10R}$$

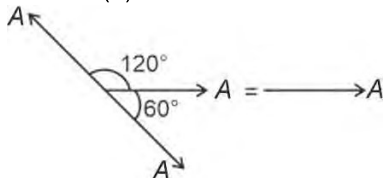
27. Answer (2)

After reflection, wave travels in negative direction.
Due to reflection from rigid boundary, a phase difference of π is introduced.

So,

$$y_r = A \sin(\omega t + kx + \pi) = -A \sin(\omega t + kx)$$

28. Answer (1)



29. Answer (2)

$$U = a + bPV$$

$$U = a + bnRT$$

$$nC_V = \frac{dU}{dT} = nbR$$

$$\Rightarrow C_V = bR$$

$$C_P = C_V + R = (b + 1)R$$

$$\gamma = \frac{C_P}{C_V} = \frac{b+1}{b}$$

30. Answer (3)

$$k_{\text{eff}} = k_1 + k_2$$

$$T = 2\pi \sqrt{\frac{m}{9k}}$$

31. Answer (1)

$$PV = nRT = \frac{m}{M}RT$$

$$\Rightarrow P \propto \rho \text{ as } T = \text{Constant}$$

32. Answer (4)

Initially gas is being compressed and later it expands.

33. Answer (3)

$y = A \sin(\omega t)$ is not an equation of stationary wave.

34. Answer (2)

$$\frac{C_P}{C_V} = \frac{n_1 C_{P_1} + n_2 C_{P_2}}{n_1 C_{V_1} + n_2 C_{V_2}}$$

$$= \frac{5 \times \frac{5R}{2} + 10 \times \frac{7R}{2}}{5 \times \frac{3R}{2} + 10 \times \frac{5R}{2}}$$

$$= \frac{25R + 70R}{15R + 50R} = \frac{95R}{65R}$$

$$\frac{C_P}{C_V} = \frac{19}{13}$$

35. Answer (4)

$$C_P - C_V = R$$

$$K_B = \frac{R}{N_A}$$

$$C_V = \frac{R}{\gamma - 1}$$

$$C_P = \frac{R\gamma}{\gamma - 1}$$

Hence the fourth relation given is wrong.

SECTION - B

36. Answer (2)

In isobaric process, $\Delta P = 0$

In isochoric process, $\Delta V = 0$

In isothermal process, $\Delta T = 0$

In adiabatic process $\Delta Q = 0$

37. Answer (3)

$$U = \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT$$

$$= \frac{5}{2} \times 4RT + \frac{3}{2} \times 6RT$$

$$= 10RT + 9RT$$

$$U = 19RT$$

38. Answer (3)

1 → 2 (Isothermal), 2 → 3 (Isobaric),

3 → 1 (Isochoric)

In process 1-2, T = C, P ↓

In process 2 → 3, V ↓ T ↓

In process 3 → 1, P ↑ T ↑

39. Answer (1)

Particle velocity (v_p) = -slope × wave velocity

40. Answer (4)

$$\frac{dx}{dt} = v = -4 \times 4\pi \sin(4\pi t)$$

$$v = -16\pi \sin 4\pi t$$

For speed to be maximum

$$\sin 4\pi t = 1$$

$$\Rightarrow 4\pi t = \frac{\pi}{2}$$

$$t = \frac{1}{8} \text{ s}$$

41. Answer (1)

$$\text{K.E} = \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t$$

$$\text{T.E} = \frac{1}{2} m \omega^2 a^2$$

$$\text{K.E} = 0.5 \text{ T.E}$$

$$0.5 = \cos^2 \omega t \Rightarrow \cos \omega t = \frac{\pm 1}{\sqrt{2}}$$

For minimum time,

$$\omega t = \frac{\pi}{4} \Rightarrow t = \frac{\pi}{4 \times \omega}$$

$$t = \frac{\pi \times 8}{4 \times 2\pi} = 1 \text{ s}$$

42. Answer (2)

$$f = \frac{nv}{2l}$$

$$180 = \frac{4v}{2 \times 4}$$

$$v = 360 \text{ m/s}$$

$$f_1 = \frac{v}{2l} = \frac{360}{2 \times 4} = 45 \text{ Hz}$$

43. Answer (3)

Time taken by wave pulse to reach support is

$$= 2\sqrt{\frac{l}{g}} = 2\sqrt{\frac{40}{10}} = 4 \text{ s}$$

44. Answer (3)

$$T_A = 2\pi\sqrt{\frac{3m}{K}}$$

$$T_B = 2\pi\sqrt{\frac{m}{3K}} = \frac{T_A}{3}$$

$$\frac{T_A}{T_B} = 3$$

45. Answer (4)

Average translational kinetic energy

$$= \frac{3}{2} K_B T$$

∴ It is independent of degree of freedom

 $\frac{C_P}{C_V}$ decreases as atomicity increases.

46. Answer (1)

$$Q = \frac{Q}{4} + W$$

$$W = \frac{3Q}{4} = \frac{R}{1-x} \Delta T$$

$$\Delta U = \frac{Q}{4} = \frac{3}{2} R \Delta T$$

$$\frac{9}{2} R = \frac{R}{1-x}$$

$$1-x = \frac{2}{9}$$

$$x = \frac{7}{9}$$

47. Answer (4)

Frequency of oscillation of kinetic energy is twice the frequency of S.H.M.

48. Answer (4)

$$W = nRT \ln \frac{V_2}{V_1}$$

$$W = nRT \ln \left(\frac{5}{2} \right)$$

$$W' = nRT \ln \left(\frac{25V}{10V} \right) = W$$

49. Answer (4)

Temperature is constant therefore internal energy of ideal gas remains constant.

50. Answer (4)

Third overtone has a frequency $7n$, which means

$$L = \frac{7\lambda}{4} = \text{three full loops} + \text{one half loop}$$

Which would make four nodes and four antinodes.

CHEMISTRY

SECTION - A

51. Answer (3)

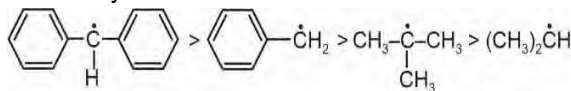
In , double bond at bridgehead carbon is highly unstable, therefore will not undergo hyperconjugation.

52. Answer (3)

Boiling point increases with increase in molecular mass. For molecules with same molecular mass, more branching decreases the boiling point.

53. Answer (4)

In $(C_6H_5)_2\dot{C}H$, maximum number of resonating structures stabilise the free radical the most. Order of stability is as follows:



54. Answer (4)

Hydrogenation of alkynes give selectively trans alkene with Na/liquid NH_3 .

55. Answer (3)

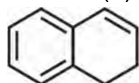
Rate of hydrogenation decreases with increasing substitution in the alkene.

56. Answer (1)

- $CH_3OC_3H_7$ & $C_2H_5OC_2H_5$
 Methoxypropane & Ethoxyethane
- Molecular formula – $C_4H_{10}O$
- Functional group – Ether

Both have same molecular formula as well as same functional group, only alkyl chains are different on either side of functional group. So, these are metamers.

57. Answer (4)

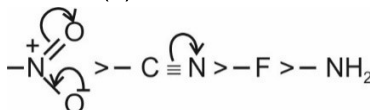


Aromatic because of benzene ring.



Non-aromatic as delocalization of π electrons does not occurs.

58. Answer (4)

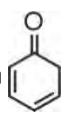
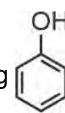


59. Answer (3)

H-I bond is weak and iodine free radical formed get combined to form iodine molecules instead of addition to double bond.

So, there is no peroxide effect. Hence, no anti Markovnikov's addition occurs here.

60. Answer (2)

Aromatization occurs in  leading  which is stable.

61. Answer (4)

Different fractions of crude oil can be separated by fractional distillation.

62. Answer (2)

Molar mass of AgBr = 188 g

$$\text{Mass of Br in } 1.88 \text{ g AgBr} = \frac{80}{188} \times 1.88 = 0.8 \text{ g}$$

$$\therefore \% \text{ of Br} = \frac{0.8}{2} \times 100 = 40\%$$

63. Answer (4)

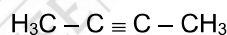
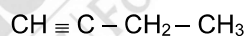
Geometrical isomerism is shown by alkenes of types $XYC = CXZ$, $XYC = CZW$ and $YXC = CXY$.

64. Answer (1)

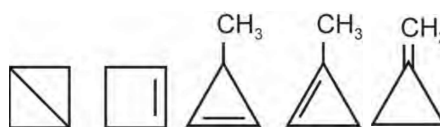
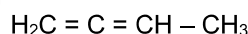
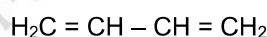
Degree of unsaturation = 2

C_4H_6

Alkynes



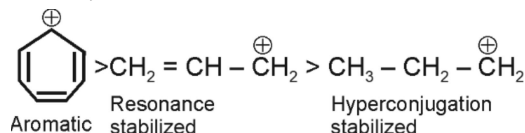
Alkenes



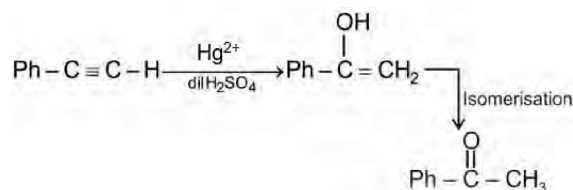
Total cyclic isomers formed are 5.

65. Answer (4)

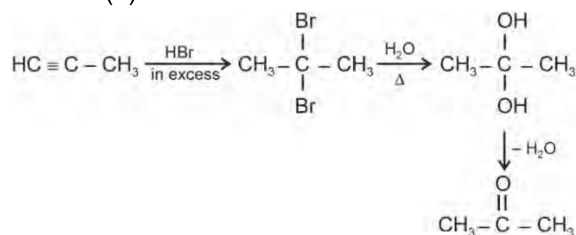
Stability of carbocation.



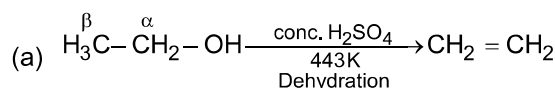
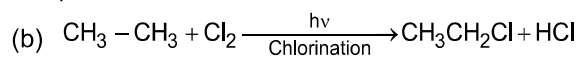
66. Answer (3)



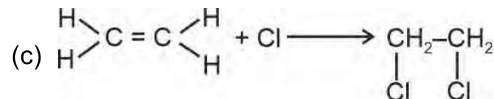
67. Answer (3)



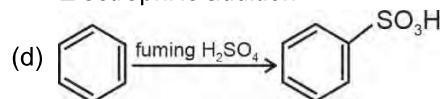
68. Answer (4)

 β -Elimination

Free radical substitution



Electrophilic addition

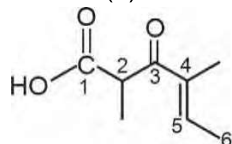


Electrophilic aromatic substitution

69. Answer (3)

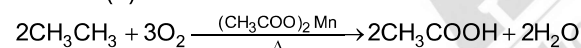
–NO₂ is electron withdrawing group and decreases reactivity towards electrophilic attack.

70. Answer (4)



2, 4-Dimethyl-3-oxohex-4-enoic acid

71. Answer (4)

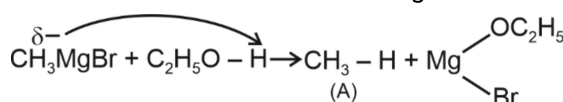


72. Answer (2)

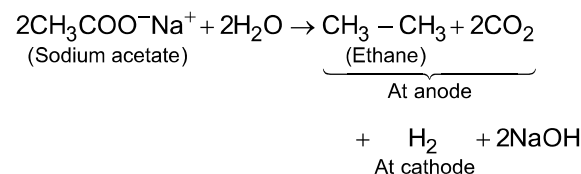
Alkanes can have infinite number of conformations by rotation around C–C single bond.

73. Answer (1)

Grignard reagent consists of $\overset{\delta-}{\text{R}} - \overset{\delta+}{\text{MgBr}}$

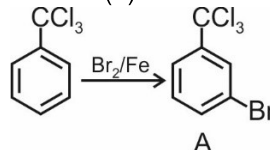


74. Answer (2)



Electrolysis of salt of carboxylic acid produces alkane with even number of carbon atom.

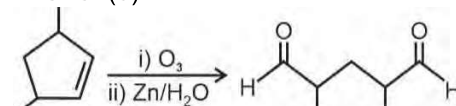
75. Answer (4)



76. Answer (1)

$\text{CH}_3 - \text{CH}_2 - \underset{\text{F}}{\underset{|}{\text{CH}}} - \overset{+}{\text{CH}_2}$ is the most stable carbanion because of –I effect of –F attached to next carbon.

77. Answer (3)

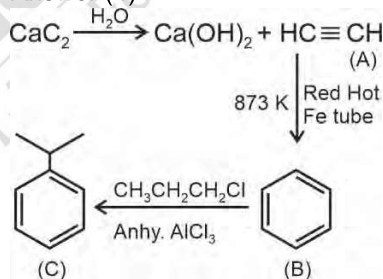


78. Answer (3)

it exist as non-planar structure out of plane whereas in all atoms are sp² hybridized so it is planar structure. sp³ carbon is present in a ring in case of

can never be coplanar.

79. Answer (4)



80. Answer (3)

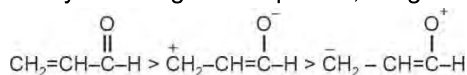
Electrophiles are electron deficient species while Lewis bases are electron rich species. So, electrophiles are not Lewis bases. :CCl₂ carbene is electron deficient species and acts as electrophile.

81. Answer (1)

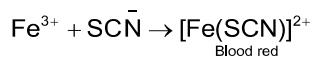
We can check the order of the stability of resonating structure by following the order

1. Structure neutrality
2. Most electronegative element contain negative charge and less electronegative element contain positive charge.

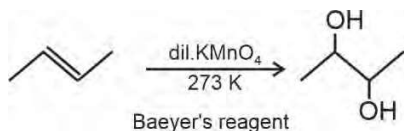
So by following this sequence, we get the order:



82. Answer (3)



83. Answer (1)



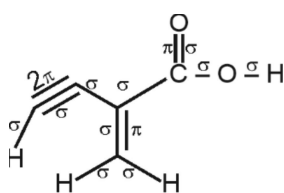
84. Answer (1)

sp^2 hybridized carbon in $\text{CH}_3 - \text{C} = \text{C} - \text{H}$ is vinylic

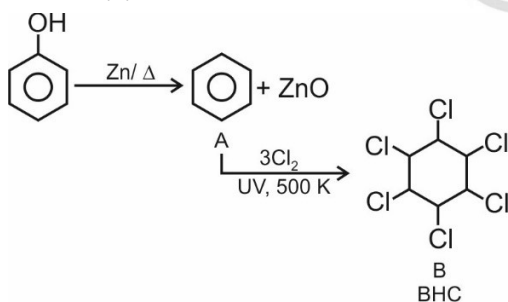
$\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{C} = \text{C} - \text{H} \end{array}$

Hence, 3 vinylic H atoms.

85. Answer (4)

 π bonds = 4 σ bonds = 10**SECTION-B**

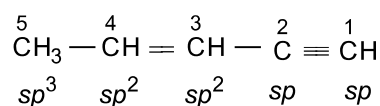
86. Answer (2)



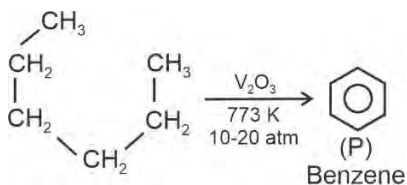
87. Answer (3)

Adsorption chromatography is based upon differential adsorption of various components of mixture on a suitable adsorbent such as silica gel or alumina.

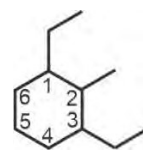
88. Answer (1)



89. Answer (4)

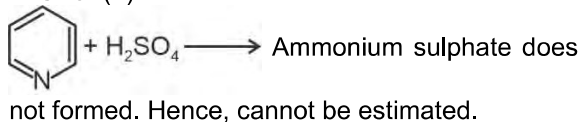


90. Answer (2)



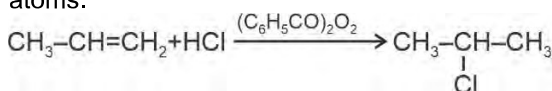
1,3-diethyl - 2 - methylcyclohexane

91. Answer (1)

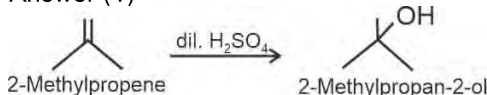


92. Answer (3)

When a reagent is added to an unsymmetrical alkene then negative half of the reagent will attach to the carbon atom containing fewer hydrogen atoms.



93. Answer (1)

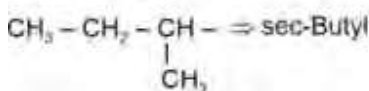
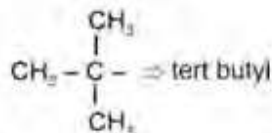
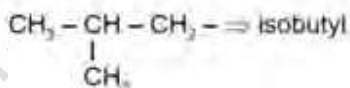


94. Answer (3)

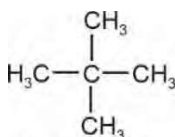
233 g BaSO_4 contains 32 g sulphur
 0.699 g BaSO_4 contains $\frac{32 \times 0.699}{233}$ g sulphur
 = 0.096

% of sulphur = $\frac{0.096 \times 100}{0.4} = 24\%$

95. Answer (3)

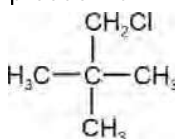
 $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - n\text{-butyl}$ 

96. Answer (2)



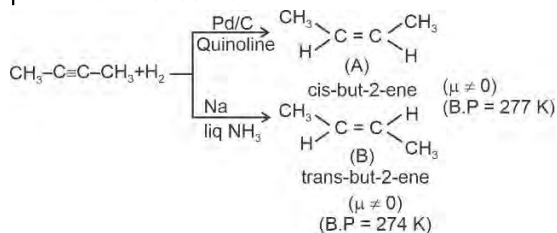
Neopentane

Monochlorination will result only in one type of product i.e.

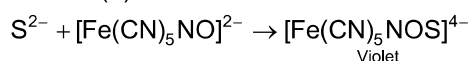


97. Answer (4)

It is observed that trans isomer has higher melting point than cis isomer.



98. Answer (1)



99. Answer (2)

$(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$ is a yellow coloured solid.

100. Answer (1)

–CHO and –COOR group show –R while –OH group exerts +R effect.

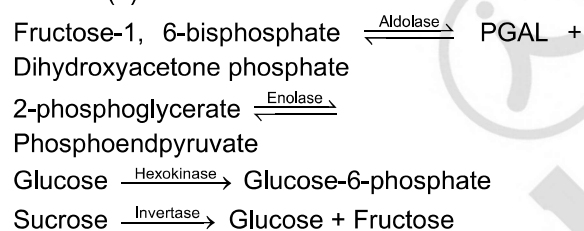
BOTANY

SECTION - A

101. Answer (3)

During non-cyclic photophosphorylation, as electrons move through the photosystems, protons are transported across the thylakoid membrane. This happens because the primary acceptor of electron which is located towards the outer side of the membrane transfers its electron not to an electron carrier but to an H carrier.

102. Answer (3)



103. Answer (2)

Pure proteins or fats are never used as respiratory substrates.

104. Answer (3)

RQ of $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) is one.

105. Answer (3)

In mitochondrial ETS, oxygen acts as the final electron acceptor.

106. Answer (2)

Pyruvic acid decarboxylase and alcohol dehydrogenase catalyse the process of alcoholic fermentation.

107. Answer (1)

Pyruvic acid is the key product of glycolysis.

108. Answer (3)

Krebs cycle takes place in mitochondrial matrix.

109. Answer (3)

Succinate dehydrogenase participates in both Krebs cycle and ETS.

110. Answer (1)

Amino acids (after deamination) depending on their structure would enter the pathway at some stage within the Krebs cycle or even as pyruvate or acetyl CoA.

111. Answer (3)

C_4 plant respond to higher temperatures and they show higher rate of photosynthesis.

112. Answer (3)

Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase. Energy is used to create a high concentration of protons within the thylakoid lumen.

During chemiosmosis, a proton gradient is created across the membrane as well as a measurable decrease in pH in the lumen occurs.

The gradient is broken down due to movement of protons across the membrane.

113. Answer (3)

Decarboxylation reaction steps in Krebs cycle are
Oxalosuccinic acid $\rightarrow$ α -ketoglutaric acid
 α -ketoglutaric acid $\rightarrow$ Succinyl CoA

114. Answer (3)

Plants breathe but they do not have specific organs for it.

115. Answer (1)

Glycolysis occurs in cytoplasm of cell in almost all organisms. It does not require oxygen.

116. Answer (2)

Pay-off phase is energy producing phase of glycolysis that involves substrate level phosphorylation.

117. Answer (3)

It is said to be floating respiration when fat/carbohydrate is used as respiratory substrate.

It is said to be protoplasmic respiration when protein is used as respiratory substrate.

118. Answer (1)

Coenzyme A is required for link reaction as well as in Krebs cycle.

119. Answer (1)

In aerobic respiration, one PGAL will produce-
2ATP via glycolysis.

1GTP (= 1ATP) via Krebs cycle

1NADH + H^+ via glycolysis

1NADH + H⁺ via link reaction
 3NADH + H⁺ and 1FADH₂ via Krebs cycle
 Net ATP = 2 + 1 + [(1 + 1 + 3) × 3] + [(1) × 2]
 = 3 + 15 + 2
 = 20 ATP

120. Answer (2)

α-ketoglutaric acid → 5 carbons
 Acetyl-coenzyme A → 2 carbons
 Dihydroxyacetone phosphate → 3 carbons
 Citric acid → 6 carbons

121. Answer (2)

Stroma lamellae membrane lacks PS II as well as NADP reductase enzyme. Oxygen evolving complex (OEC) is associated with PS II.

122. Answer (3)

Stroma lamellae is a possible location for cyclic photophosphorylation.

123. Answer (4)

Z-scheme represents non-cyclic photophosphorylation. RuBisCO participates in dark reaction of photosynthesis.

124. Answer (4)

Plastoquinone transfers the electron between PS II and cytochrome B₆f.

125. Answer (3)

Water acts as the external source of electrons for PS II.

126. Answer (2)

The action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a.

127. Answer (1)

Chlorophyll b imparts yellow-green colour in the chromatogram.

128. Answer (4)

Light reactions or the photochemical phase of photosynthesis include light absorption, water splitting, oxygen release, and the formation of high-energy chemical intermediates, ATP and NADPH.

129. Answer (1)

The use of radioactive ¹⁴C by Melvin Calvin in algal photosynthesis studies led to the discovery that the first CO₂ fixation product was a 3-carbon organic acid.

130. Answer (4)

'Kranz' anatomy is exhibited by C₄ plants. Tomato is a C₃ plant.

131. Answer (2)

Organic acids like OAA and malic acids present in C₄ plants help them to tolerate saline conditions as it lowers their water potential than that of soil.

132. Answer (2)

Photorespiration does not involve formation of ATP or NADPH but there is release of fixed CO₂ with the utilisation of ATP and thus considered as a wasteful process.

133. Answer (4)

In C₄ plants, process of carbon fixation is faster.

134. Answer (2)

Photosynthesis occurs in some prokaryotes only.

135. Answer (3)

C₃ plants show CO₂ fertilisation effect.

SECTION - B

136. Answer (4)

- | | |
|-----------------------|---|
| a. Joseph Priestley | Revealed the essential role of air in the growth of green plants |
| b. Jan Ingenhousz | Showed green parts of plant could release oxygen |
| c. Cornelius Van Niel | Demonstrated that hydrogen from a suitable oxidisable compound reduce CO ₂ to carbohydrates. |
| d. T.W. Engelmann | Described first action spectrum of photosynthesis |

137. Answer (3)

The TCA cycle requires the continued regeneration of NAD⁺ and FAD⁺ from NADH and FADH₂ respectively. There are three points in the TCA cycle where NAD⁺ is reduced to NADH + H⁺ and one point where FAD⁺ is reduced to FADH₂.

138. Answer (2)

Moll's half leaf experiment is used to demonstrate that CO₂ is necessary for photosynthesis.

139. Answer (3)

Usually the chloroplasts align themselves along the walls of the mesophyll cells, such that they get the optimum quantity of the incident light.

140. Answer (1)

Net gain of ATP is 2 when one molecule of glucose undergoes glycolysis.

141. Answer (1)

RuBisCO has a much greater affinity for CO₂ than O₂ when CO₂ : O₂ is nearly equal.

Increase in intracellular CO₂ concentration minimizes the oxygenase activity of RuBisCO and maximises carboxylase activity.

Binding of O₂ and CO₂ to RuBisCO is competitive.

142. Answer (2)

ATP synthesis via ATP synthase requires a proton gradient which is developed by proton pump. ATP is produced by facilitated diffusion of proton via ATP synthase.

143. Answer (4)

Assumptions for making respiratory balance sheet are not really valid in living system. For making the balance sheet, it is assumed that, all pathways work simultaneously and no other alternative substrate can enter in the pathway at any of the intermediary stage. Also none of the intermediate can be withdrawn from the respiratory pathway.

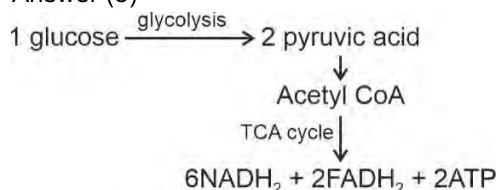
144. Answer (2)

Redox equivalents are removed during conversion of PGAL to 1, 3 BPGA resulting in formation of NADH_2 .

145. Answer (4)

In first step of TCA cycle, acetyl CoA combines with OAA and yields citric acid. This reaction is catalysed by citrate synthase.

146. Answer (3)



147. Answer (1)

Cytochrome b₆f complex accept the electrons from PS II and transfer them to PS I.

148. Answer (1)

2 molecules of ATP and 2 molecules of NADPH are consumed per CO_2 molecule fixation during the reduction step of Calvin cycle.

149. Answer (1)

Most crucial step of Calvin cycle is 1st step *i.e.*, carboxylation by RuBP and the formation of 3-PGA.

150. Answer (3)

Increase in incident light beyond a point cause the breakdown of chlorophyll and a decrease in photosynthesis.

ZOOLOGY

SECTION - A

151. Answer (1)

Reptiles, birds, land snails and insects excrete nitrogenous wastes as uric acid in the form of pellet or paste with a minimum loss of water and are called uricotelic animals.

152. Answer (2)

PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption. Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by this segment.

153. Answer (4)

Many bony fishes, aquatic amphibians and aquatic insects are ammonotelic in nature. Mammals, many terrestrial amphibians and marine fishes mainly excrete urea and are called ureotelic animals.

Ammonia is the most toxic nitrogenous waste.

154. Answer (2)

The renal tubule begins with a double walled cup-like structure called Bowman's capsule, which encloses the glomerulus. Glomerulus along with Bowman's capsule, is called the Malpighian body or renal corpuscle.

The DCTs of many nephrons open into a straight tube called collecting duct, many of which converge and open into the renal pelvis through medullary pyramids in the calyces.

155. Answer (3)

Presence of juxtamedullary nephrons is extremely important for water conservation in mammals. Processing of filtrate in the proximal and distal tubules maintains a constant pH in body fluids.

156. Answer (1)

The cranium is formed by eight bones, the paired parietal and temporal bones plus the unpaired frontal, occipital, sphenoid and ethmoid bones.

157. Answer (4)

First seven pairs of ribs are called true ribs (vertebrosternal ribs). The vertebral column is differentiated into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull. The total number of carpals and tarsals in one forelimb and hindlimb are 8 and 7 respectively.

158. Answer (4)

The thick filaments in the 'A' band are held together in the middle of this band by a thin fibrous membrane called 'M' line. In the centre of each 'I' band is an elastic fibre called 'Z' line which bisects it. The thin filaments are firmly attached to the 'Z' line.

159. Answer (3)

Skeletal muscles are primarily involved in the locomotory actions and changes of body postures while visceral muscles assist in the transportation of food through the digestive tract and gametes through the genital tract. Both skeletal and visceral muscle fibres are unbranched.

160. Answer (2)
In *Paramoecium*, cilia helps in the movement of food through cytopharynx and in locomotion as well. *Hydra* can use its tentacles for capturing its prey and also use them for locomotion. Passage of ova through the female reproductive tract is also facilitated by the ciliary movement.
161. Answer (2)
The medulla of the brain is connected to the spinal cord. The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions.
162. Answer (3)
Collections of RER and ribosomes are named as Nissl's granules. The cell body contains cytoplasm with typical cell organelles and certain granular bodies called Nissl's granules. Short fibres which branch repeatedly and project out of the cell body also contain Nissl's granules and are called dendrites.
163. Answer (1)
The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called the limbic lobe or limbic system. Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions (e.g., excitement, pleasure, rage and fear) and motivation. Centres present in medulla oblongata regulate gastric secretions and cardiovascular reflexes.
164. Answer (2)
The neural system of all animals is composed of highly specialised cells called neurons which can detect, receive and transmit different kinds of stimuli. The neural organisation is very simple in lower invertebrates. For example, in *Hydra* it is composed of a network of neurons. The neural system is better organised in insects, where a brain is present along with a number of ganglia and neural tissues.
165. Answer (3)
The PNS is divided into two divisions called somatic neural system and autonomic neural system. Visceral nervous system is the part of the peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia and plexuses.
166. Answer (3)
A comparison of the volume of the filtrate formed per day (180 litres per day) with that of the urine released (1.5 litres), suggest that nearly 99 per cent of the filtrate has to be reabsorbed by the renal tubules.
167. Answer (1)
The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries. A minute vessel of this network runs parallel to the Henle's loop forming a 'U' shaped vasa recta. Vasa recta is absent or highly reduced in cortical nephrons.
168. Answer (2)
Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acid, etc. Liver, the largest gland in our body, secretes bile-containing substances like bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins and drugs.
169. Answer (4)
A fall in glomerular blood flow/glomerular blood pressure/GFR can activate the JG cells to release renin which converts angiotensinogen in blood to angiotensin I and further to angiotensin II. Angiotensin II, being a powerful vasoconstrictor, increases the glomerular blood pressure and thereby GFR. Angiotensin II also activates the adrenal cortex to release aldosterone. Aldosterone causes reabsorption of Na⁺ and water from the distal parts of the tubule. This also leads to an increase in blood pressure and GFR.
170. Answer (2)
During shortening of the muscle, i.e., contraction, the 'I' bands get reduced, whereas the 'A' bands retain the length.
171. Answer (2)
Gliding joint is present between the carpals and saddle joint is present between carpal and metacarpal of human thumb.
172. Answer (1)
Osteoporosis: Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.
173. Answer (3)
The gaps between two adjacent myelin sheaths are called nodes of Ranvier.
174. Answer (3)
The cerebral cortex is referred to as the grey matter due to its greyish appearance. The neuron cell bodies are concentrated here giving the colour.
175. Answer (3)
The ionic gradients across the resting membrane are maintained by the active transport of ions by the sodium-potassium pump which transports 3 Na⁺ outwards for 2 K⁺ into the cell on expenditure of one ATP.
176. Answer (1)
PCT is internally lined by simple cuboidal brush bordered epithelium which increases the surface area for reabsorption. Bowman's capsule is lined by simple squamous epithelium.

177. Answer (3)

JGA is formed by cellular modifications in DCT and afferent arteriole at the location of their contact.

178. Answer (4)

On an average, 1100-1200 mL of blood is filtered by the kidneys per minute which constitutes roughly $\frac{1}{5}$ th of the blood pumped out by each ventricle of heart in a minute. So blood filtered per hour is

$$1100 \times 60 = 66000 \text{ mL or } 1200 \times 60 = 72000 \text{ mL} \\ = 66000 - 72000 \text{ mL}$$

179. Answer (1)

ADH is anti-diuretic hormone.

Alcohol and caffeine are diuretics.

180. Answer (3)

Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called uremia, which is highly harmful and may lead to kidney failure.

181. Answer (4)

Muscles have special properties like excitability, contractility, extensibility and elasticity.

182. Answer (1)

Hyoid is a single U-shaped bone present at the base of the buccal cavity and is included in the skull.

183. Answer (2)

The bone has a very hard matrix due to calcium salts and cartilage has slightly pliable matrix due to chondroitin salts.

184. Answer (4)

White muscle fibres exhibit high number of sarcoplasmic reticulum.

185. Answer (3)

LMM and HMM are components of meromyosin.

SECTION - B

186. Answer (2)

Vestibular apparatus is responsible for maintenance of balance of the body and posture.

187. Answer (1)

All ribs are connected dorsally to vertebral column.

188. Answer (3)

Chemical integration is provided through hormones. Both neural system and endocrine system of the body control various functions of the body. Coordination is the process through which two or more organs interact and complement the functions of each other.

189. Answer (3)

Pia mater is innermost meninx of brain and remains in direct contact with the brain tissue.

190. Answer (1)

Cerebrum wraps around a structure called thalamus, which is major coordinating centre for sensory and motor signalling.

191. Answer (1)

ATP is hydrolysed releasing the energy during muscle contraction.

192. Answer (4)

Micturition is initiated by the stretch receptors in the urinary bladder. Sympathetic stimulation causes the constriction of internal urethral sphincter and relaxation of urinary bladder muscles. Parasympathetic stimulation causes contraction of urinary bladder muscles and relaxation of internal urethral sphincter.

193. Answer (3)

Involuntary muscles are present in hollow organs such as intestine.

194. Answer (2)

Selective reabsorption of urea occurs in CD.

195. Answer (3)

TpC is calcium binding subunit on troponin.

196. Answer (4)

Kidneys are bean shaped structures situated between the levels of T₁₂ to L₃.

197. Answer (2)

Trachea in insects and book lungs in scorpion are considered as respiratory structures.

198. Answer (3)

At a chemical synapse, the membranes of the pre- and post-synaptic neurons are separated by a fluid-filled space called synaptic cleft. Chemicals called neurotransmitters are involved in the transmission of impulses at these synapses.

199. Answer (2)

Bipolar neurons possess cell bodies with one axon and one dendrite.

200. Answer (2)

Coordination is the process through which two or more organs interact and complement the functions of one another.

Neurological cells are the constituents of nervous system.